
[17] - Medicine Ordering System

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Introduction:

The Medicine Ordering System (MOS) is used to help patients safely and efficiently order prescribed and over-the-counter medication online. The system acts as a platform where patients can browse pharmaceutical products, upload their prescriptions, and track their ordered medication. It will connect to external hospital e-prescription databases for automated verification and third-party payment gateways for secure transactions. The system fits into the strategic objectives of the commissioning pharmacy by increasing medication accessibility, expanding sales reach beyond the physical storefront, and reducing patient wait times in-store.

User Requirements:

1. As a patient, I want to securely upload a picture of my doctor's prescription so that the pharmacist can prepare my medication.
2. As a patient, I want to receive alerts when my chronic medication is running low so I don't miss a dose.
3. As a patient, I want the system to verify my uploaded prescriptions against other records to ensure accurate dispensing and my safety.
4. As a patient, I want to see a real-time map of my delivery driver so I know exactly when my medicine will arrive.
5. As a delivery driver, I want the system to optimize my delivery route so I can drop off medications efficiently.
6. As a pharmacist, I want to check medical interactions of prescribed medications to ensure patient safety.

Functional Requirements (Ranked by Priority):

1. Name: User Login

- **Description/Action:** The system authenticates users by verifying their credentials and grants access to their respective dashboards.
- **Requirements/Inputs:** Username or email password.
- **Source:** User login interface.
- **Pre-condition:** The user has a registered account .The user is not already logged in.
- **Post-condition:** If credentials are valid → user is successfully logged in and redirected to their dashboard. If invalid → access is denied and an error message is displayed.
- **Output:** Successful login → dashboard view. Failed login → error notification ("Invalid email or password").

2. Name: Cart

- **Description/Action:** Allows the patient to add over-the-counter medications to their shopping cart before placing the order. If the selected medication requires a prescription, allow the patient to add medications to their shopping cart after prescription verification before placing the order.
- **Requirements/Inputs:** Medication ID, quantity, patient ID.
- **Source:** Medicine catalog interface.
- **Precondition:** Patient is logged in. Selected medication is available and requires verification if a prescription is uploaded.
- **Post-condition:** Selected medication is stored temporarily in the patient's cart.
- **Output:** Confirmation message ("Item added to cart successfully") and updated cart contents.

3. Name: Place Order

- **Description/Action:** Allows the patient to submit a medicine order.
- **Requirements/Inputs:** Medications, quantities, patient ID, delivery address
- **Source:** Checkout interface.
- **Pre-condition:** Patient is logged in, and the cart is not empty.
- **Post-condition:** Order is placed, and payment is completed.
- **Output:** Order confirmed, and a message is displayed to the user.

4. Name: Prescription Verification

- **Description/Action:** The system reviews, validates, and either approves or rejects uploaded prescriptions.
- **Requirements/Inputs:** Uploaded prescription image file, patient ID.
- **Source:** Patient account.
- **Pre-condition:** A patient has uploaded a prescription.
- **Post-condition:** The order status is updated to 'Approved' (ready for payment/dispatch) or 'Rejected' (canceled).
- **Output:** An approval or rejection notification is sent to the patient's device.

5. Name: Secure Payment Processing

- **Description/Action:** Handles the financial transaction for the medicine order using integrated third-party gateways.
- **Requirements/Inputs:** Cart total amount, payment method details, authentication token.
- **Source:** Patient checkout interface.
- **Pre-condition:** The cart is finalized, and any required prescriptions have been verified.
- **Post-condition:** Funds are deducted from the user's account, and the order is queued for fulfillment.
- **Output:** A transaction success screen and an electronic PDF receipt sent via email.

6. Name: Drug Interaction Checking

- **Description/Action:** The system cross-references newly ordered medications against the patient's active prescriptions to flag potential adverse health interactions.
- **Requirements/Inputs:** New medication ID, patient's active medication list.
- **Source:** National Drug Database/Pharmacy Inventory System.
- **Pre-condition:** A patient adds a new medication to their cart.
- **Post-condition:** The interaction risk level is assessed and logged.
- **Output:** A warning alert is displayed on the patient's screen and flagged for the reviewing pharmacist.

7. Name: Prescription Upload

- **Description/Action:** Allows the patient to upload a prescription image to the system.
- **Requirements/Inputs:** Image file, patient ID.
- **Source:** Upload Interface.
- **Pre-condition:** Patient is logged in.
- **Post-condition:** Prescription is stored and linked to an order.
- **Output:** Confirmation message.

8. Name: Real-time Order Tracking

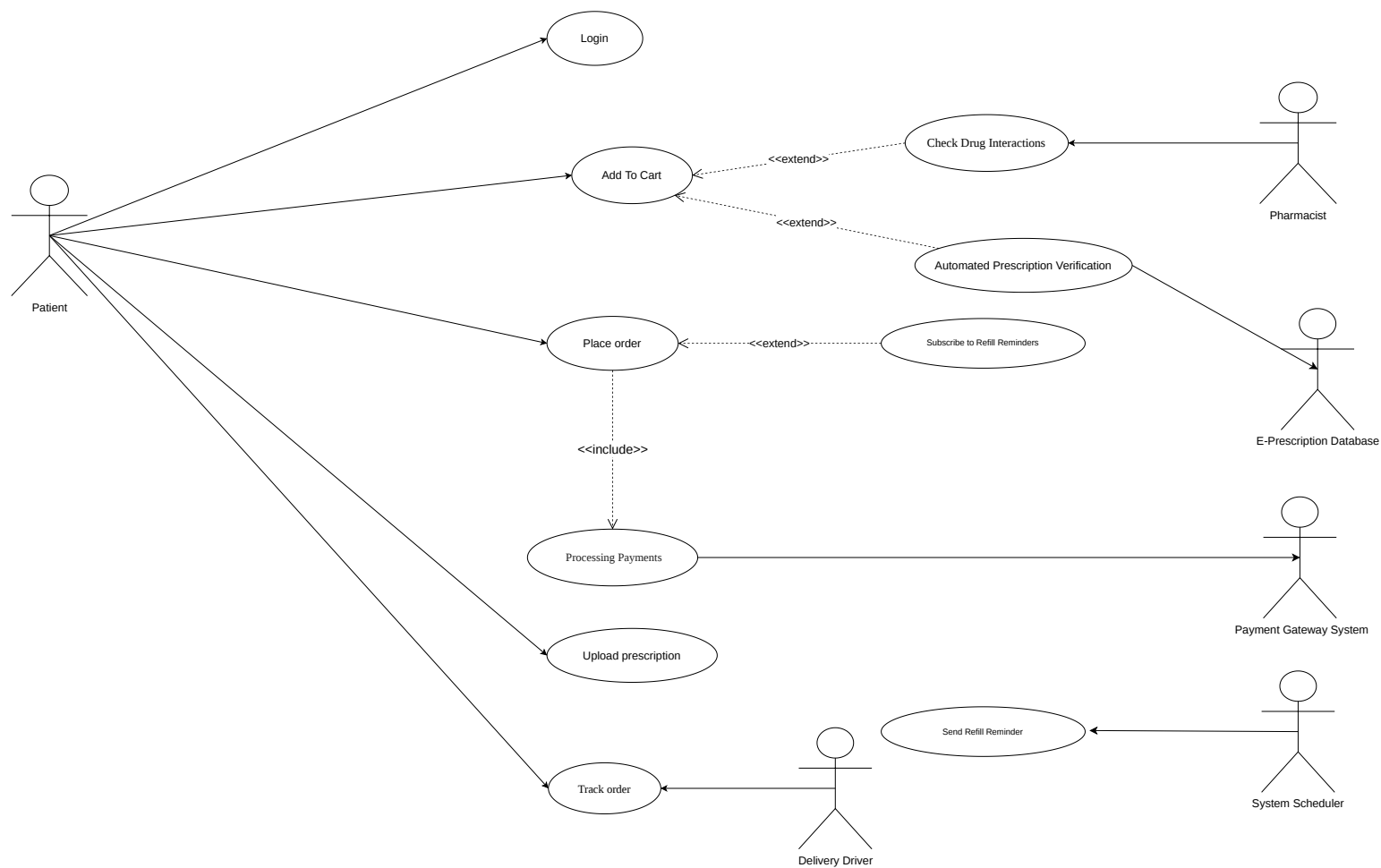
- **Description/Action:** Calculates and displays the current location and estimated time of arrival of the assigned delivery driver.
- **Requirements/Inputs:** Order ID, live GPS coordinates of the delivery driver.
- **Source:** Delivery driver mobile app GPS module.
- **Pre-condition:** The verified order has been dispatched from the pharmacy.
- **Post-condition:** The backend routing cache is updated with the latest coordinates.
- **Output:** A live map view and continuously updating estimated time of arrival timestamp on the patient's app.

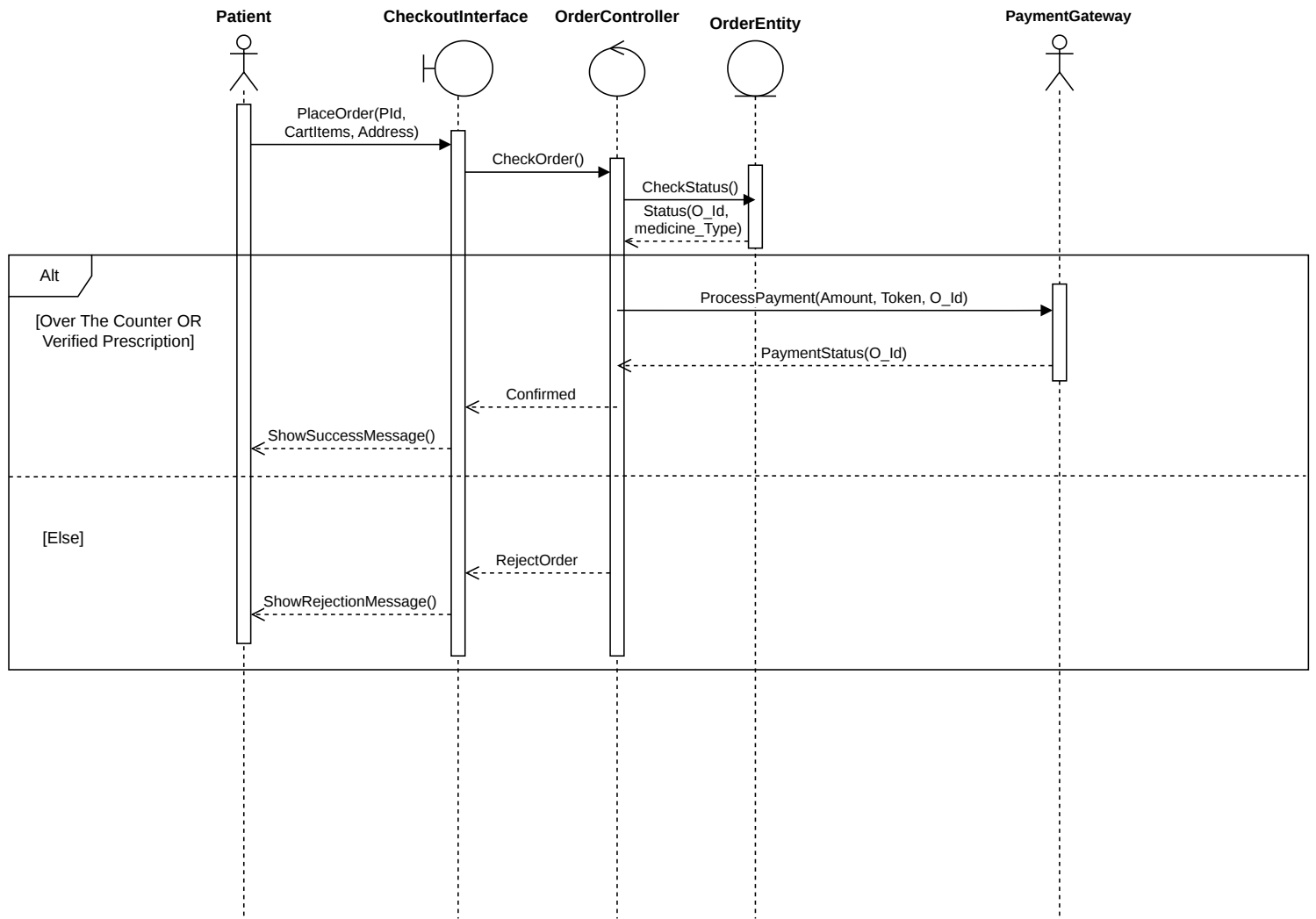
9. Name: Automated Refill Reminders

- **Description/Action:** The system calculates when a patient's chronic medication supply will run out based on their dosage and sends a proactive reminder.
- **Requirements/Inputs:** Medication dosage frequency, quantity supplied, last purchase date.
- **Source:** Patient medical profile and purchase history database.
- **Pre-condition:** The patient is subscribed to chronic medication refill tracking.
- **Post-condition:** A refill trigger is logged into the communication queue.
- **Output:** An automated push notification and email reminding the patient to reorder.

Non-Functional Requirements:

1. **Reliability:** The system must guarantee 99.9% uptime to ensure patients can access emergency or critical medication orders at any time without service interruption.
2. **Performance (Response Time):** The catalog search and filtering queries must return results to the user interface in under 2.0 seconds to ensure a smooth browsing experience.
3. **Security (Constraints):** All patient data, including prescription images and medical history, must be encrypted at rest and in transit in strict compliance with healthcare data privacy laws.
4. **Development Method & Language (System Representations):** The database will be managed by SQL developer, and the application logic will be developed using C#.NET framework.
5. **Usability (Interface Requirements):** The mobile user interface must support dynamic text sizing and screen readers to accommodate elderly patients or users with visual impairments.





Use Case: Place Medication Order

Primary Actor: Patient

Precondition: Patient is logged into the system and has a valid prescription available whether it is uploaded or in the E-Prescription System.

1. Primary Flow

1. **Select Medication:** The Patient searches for medication or uploads a prescription and triggers the Add to Cart use case.
2. **Verify Prescription:** if the patient uploads a prescription, the system automatically triggers the Verify prescription extension by communicating with the E-Prescription System to ensure the medication matches the doctor's orders.
3. **Check Interactions:** Simultaneously, the Pharmacist performs the Check Drug Interactions extension to ensure patient safety.
4. **Initiate Checkout:** The Patient selects Place order.
5. **Process Payment :** The system automatically calls the Processing Payments use case.
 - The system sends a request to the Payment Gateway System.
 - The Payment Gateway returns a "Success" confirmation.
6. **Confirm Order:** The system confirms the order and provides a tracking number.
7. **Order Fulfillment:** The Delivery Driver updates the status, allowing the Patient to track their order.

2. Alternate Flows

A. Uploading a Manual Prescription

1. If the medication is not found in the E-Prescription System, the Patient chooses the Upload prescription option.
2. The Patient uploads a photo of the physical script.
3. The system automatically verifies the document before the order can proceed to "Place order".

B. Subscription for Refills

1. During the Place order phase, patients with chronic diseases and take medications regularly can choose to Subscribe to Refill Reminders.
 2. The system records the dosage duration.
 3. The System Scheduler triggers the Send Refill Reminder use case before the medication finishes.
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3. Extension Points and Relationships

Relationship Type	Base Use Case	Extended/Included Use Case	Trigger Condition
Include	Place order	Processing Payments	Mandatory payment must occur to complete the order.
Extend	Add to Cart	Check Drug Interactions	Occurs if the patient has other active medications on file.
Extend	Add to Cart	Verify prescription	Occurs to validate the legality of the dispense.
Extend	Place order	Subscribe to Refill Reminders	Optional triggered if the patient opts for chronic medication.

4. Post-Conditions

- **Success:** The order is logged, payment is captured, and the delivery process is initiated.
- **Failure:** If payment fails or prescription verification is rejected, the order remains in "Pending" or "Cancelled" status, and the Patient is notified to place another order.